

### Challenge (Habanero)

- Q1.** To rationalize the denominator of  $\frac{1}{\sqrt{3}}$ , we multiply by:
- (A)  $\frac{\sqrt{3}}{\sqrt{3}}$   
(B)  $\frac{2}{2}$   
(C)  $\frac{3}{3}$   
(D)  $\frac{\sqrt{2}}{\sqrt{2}}$   
(E)  $\frac{\sqrt{6}}{\sqrt{6}}$
- Q2.** A chef needs  $\frac{1}{\sqrt{2}}$  cups of flour for a recipe. To rationalize the denominator, the chef should multiply by:
- (A)  $\frac{\sqrt{2}}{\sqrt{2}}$   
(B)  $\frac{2}{2}$   
(C)  $\frac{3}{3}$   
(D)  $\frac{\sqrt{3}}{\sqrt{3}}$   
(E)  $\frac{\sqrt{6}}{\sqrt{6}}$
- Q3.** A builder needs to rationalize the denominator of  $\frac{2}{\sqrt{5}}$  to calculate the length of a beam. The rationalized form is:
- (A)  $\frac{2\sqrt{5}}{5}$   
(B)  $\frac{2\sqrt{3}}{5}$   
(C)  $\frac{4}{\sqrt{5}}$   
(D)  $\frac{2}{5}$   
(E)  $\frac{\sqrt{5}}{2}$
- Q4.** A recipe calls for  $\frac{3}{\sqrt{6}}$  teaspoons of spice. To rationalize the denominator, we get:
- (A)  $\frac{3\sqrt{6}}{6}$   
(B)  $\frac{3\sqrt{2}}{2}$   
(C)  $\frac{3\sqrt{3}}{3}$   
(D)  $\frac{3}{6}$   
(E)  $\frac{\sqrt{6}}{3}$
- Q5.** The height of a building is  $\frac{10}{\sqrt{2}}$  meters. The rationalized form of this height is:
- (A)  $\frac{10\sqrt{2}}{2}$   
(B)  $\frac{10}{2}$   
(C)  $\frac{10\sqrt{3}}{3}$   
(D)  $\frac{5}{\sqrt{2}}$   
(E)  $\frac{5\sqrt{2}}{2}$
- Q6.** To rationalize the denominator of  $\frac{2\sqrt{3}}{\sqrt{6}}$ , we multiply by:
- (A)  $\frac{\sqrt{6}}{\sqrt{6}}$   
(B)  $\frac{\sqrt{2}}{\sqrt{2}}$   
(C)  $\frac{\sqrt{3}}{\sqrt{3}}$   
(D)  $\frac{2}{2}$   
(E)  $\frac{3}{3}$
- Q7.** The volume of a rectangular prism is  $\frac{12}{\sqrt{3}}$  cubic meters. The rationalized form of this volume is:
- (A)  $\frac{12\sqrt{3}}{3}$   
(B)  $\frac{12}{3}$   
(C)  $\frac{12\sqrt{2}}{2}$   
(D)  $\frac{4}{\sqrt{3}}$   
(E)  $\frac{4\sqrt{3}}{3}$
- Q8.** A recipe for a cake calls for  $\frac{2}{\sqrt{8}}$  cups of sugar. To rationalize the denominator, we get:
- (A)  $\frac{2\sqrt{8}}{8}$   
(B)  $\frac{2\sqrt{2}}{4}$   
(C)  $\frac{\sqrt{2}}{2}$   
(D)  $\frac{1}{\sqrt{2}}$   
(E)  $\frac{1}{2}$

#### Open Ended Questions (FRQ)

- Q9.** Rationalize the denominator of  $\frac{3\sqrt{2}}{\sqrt{6}}$  and simplify. Show all steps.
- Q10.** A construction company needs to rationalize the denominator of  $\frac{5}{\sqrt{10}}$  to calculate the length of a bridge. Rationalize the denominator and simplify. Show all steps.

#### Chef's Tips

- Read each question carefully before answering.
- Show all work and simplify answers when necessary.
- Use a calculator only when instructed to do so.